

Q1:)

①

A 3-stage CMOS ring oscillator is implemented using identical inverters. Each inverter drives a load capacitance of $C_L = 20 \text{ fF}$.

The active transistor provides an approximately constant saturation current that depends on the supply voltage as follows:

for $V_{DD} = 1.0 \text{ V}$

$$I_{\text{sat}} = 50 \text{ } \mu\text{A}$$

for $V_{DD} = 1.5 \text{ V}$

$$I_{\text{sat}} = 120 \text{ } \mu\text{A}$$

Assumption:

Transistor currents assumed to remain constant (although it is not). Same current for charge and discharge.

The inverter is approximated by

$$t_p = (t_{pH} + t_{pL})/2$$

$$t_p \approx \frac{C_L V_{DD}}{I_{\text{sat}}}$$

Propagation delay is defined at the 50% output voltage level.

a) calculate the inverter delay for both supply voltages.

$$V_{DD} = 1.0 \text{ V}$$

$$t_p = \frac{20 \cdot 10^{-15} \cdot 1.0}{50 \cdot 10^{-6}} = \underline{\underline{400 \text{ ps}}}$$

$$V_{DD} = 1.5V$$

(2)

$$t_p = \frac{20 \cdot 10^{-15} \cdot 1.5}{120 \cdot 10^{-6}} = \underline{250ps}$$

b) calculate the ring oscillator frequency for both cases.

$$f = \frac{1}{2N t_p}, N=3$$

$$V_{DD} = 1.0V$$

$$f = \frac{1}{2 \cdot 3 \cdot 400 \cdot 10^{-12}} = 415MHz$$

$$V_{DD} = 1.5V$$

$$f = \frac{1}{2 \cdot 3 \cdot 250 \cdot 10^{-12}} = 667MHz$$

c) Briefly explain why the oscillator frequency increases when V_{DD} increases.

* when the supply voltage increases:

- transistors have higher overdrive voltages:

$$V_{GS} - V_{th} \downarrow$$

- the saturation current increases

- the inverter delay decreases

Q2:)

3

A 12-bit SAR ADC uses a DAC and is designed to operate with an ideal reference voltage of

$$V_{\text{ref, ideal}} = 1.2 \text{ V}$$

However, due to reference buffer inaccuracies, the actual reference voltage is measured as

$$V_{\text{ref, actual}} = 1.212 \text{ V}$$

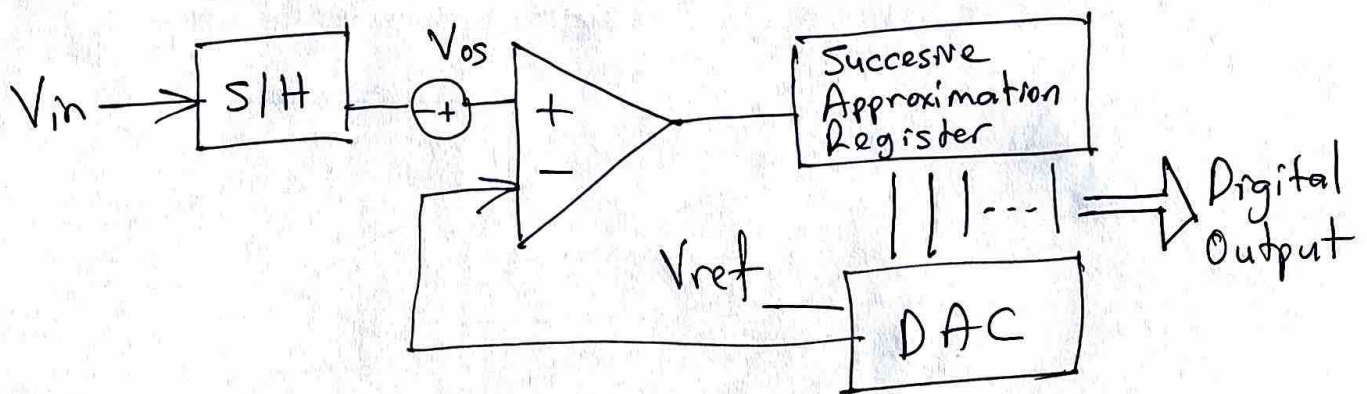
The input voltage is

$$V_{\text{in}} = 0.742 \text{ V}$$

The ADC also has a comparator offset of

$$V_{\text{os}} = +0.5 \text{ LSB}$$

Assume: MSB-first SAR operation



a) Calculate the ideal LSB size

$$\text{LSB}_{\text{ideal}} = \frac{V_{\text{ref, ideal}}}{2^{12}} = \frac{1.2}{4096} = \underline{0.293 \text{ mV}}$$

b) Determine the ideal output code without any errors included!

$$\text{Ideal Code} = \frac{0.742}{0.000293} = \underline{\underline{2532}}$$

(4)

c) Calculate the gain error (%) caused by the reference deviation.

$$\text{LSB}_{\text{dev}} = \frac{1.212}{4096} = 0.0002959 \text{ V}$$

$$\text{LSB}_{\text{dev}} = \underline{\underline{0.296 \text{ mV}}}$$

If V_{measured} is calculated in SW and assumed that $V_{\text{ref}} = 1.2 \text{ V}$!

$$V_{\text{measured}} = 2507 \cdot \frac{1.2}{4096}$$

$$\frac{V_{\text{in}}}{V_{\text{ref, actual}}} = \frac{0.742}{1.212} = 0.612$$

$$\nearrow \underline{\underline{V_{\text{measured}} = 0.734 \text{ V}}}$$

$$\text{Actual Code} = 0.612 \cdot 4096 = \underline{\underline{2507}}$$

$$\text{Gain Error} = \frac{1.212 - 1.2}{1.2} = 0.01 \rightarrow \underline{\underline{-1\%}}$$

d) Find the final output code considering both gain error and comparator offset.

$$\text{Code}_{\text{final}} = 2507 + 0.5 \approx \underline{\underline{2508}}$$

$$V_{\text{OS}} = 0.5 \times \text{LSB} \rightarrow \underline{\underline{0.148 \text{ mV}}}$$

LSB voltage

this approach is wrong! cannot be directly added! $\rightarrow$ gain error!

$$V_{\text{in, measured}} = 0.742 + 0.000148 = \underline{\underline{0.742148 \text{ V}}}$$

$$\text{Code} = \frac{0.742148}{1.212} \cdot 4096 \approx \underline{\underline{2508 \text{ LSBs}}}$$